

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)
FACULTY OF MANAGEMENT STUDIES (FMS)
BACHELOR OF BUSINESS ADMINISTRATION (BBA PROGRAMME)
SEMESTER – IV, UNIVERSITY EXAMINATION, MAY 2018
BM208.1 OPERATIONS RESEARCH

Date: 03/05/2018, Thursday

Time: 10.00 AM to 10.30 AM

Total Marks: 15

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Important Instructions:

- All questions are compulsory.
- This is a closed book examination.
- There are two sections in the question paper. Section – I contains Q. No.: 1 which is of 15 marks and Section – II contains total five questions wherein Q. No.: 2 to 5 are of 10 marks each and Q. 6 is a Case study of 15 marks.
- Section – I should be answered in the given question paper and Section – II should be answered in a separate answer book.
- Maximum time allotted for Section – I is 30 minutes. Answer Book for Section – II will be given when the Question Paper for Section – I is returned. However, candidates who complete Section – I earlier than 30 minutes, can collect Answer Book for Section – II immediately after handing over Section – I question paper.
- Figures on the right indicate marks for each question.

SECTION – I

Q.1 Objective Type Questions:

A) Match the following: (i – vi)

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Q. No.:		Probable Answer Keys		Write Your Answer Here
i.	Simplex Method	a)	Hungarian Method	i. -
ii.	Graphical Method	b)	Exactly Two Decision Variables	ii. -
iii.	Two – Person Zero-Sum Game	c)	More than Two Decision Variables	iii. -
iv.	Assignment Problem	d)	Gain of one is loss of other	iv. -
v.	Critical Activity	e)	Penalty Based Assignment	v. -
vi.	VAM	f)	Total Float = 0	vi. -
		g)	Free Float = 1	
		h)	Lowest Cost Assignment	
		i)	Laplace Theorem	
		j)	Operational Models	

B) Multiple-Choice Questions (vii – ix) (Note: Encircle the correct answer)

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vii. Initial feasible solution of transportation problem can be identified using

a) NWCR

- b) PERT
- c) Hungarian
- d) None of these
- viii.** CPM ends with
 - a) Forward Pass
 - b) Backward Pass
 - c) Floating
 - d) None of these
- ix.** Prime objective of transportation problem is to
 - a) Minimize the Cost
 - b) Maximize Profit
 - c) Balance Profit and Loss
 - d) None of these

C) Fill-in the Blanks (x – xii) You may choose from the following answers:

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Added	Subtracted	Free Float	Critical Path	Total Float	PERT / CPM
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- x.** Slack Variables are to be _____ when the constraint are of grater or equal to type.
- xi.** An activity can have delay up to certain limit so that entire project will not going to suffer and that limit can be determined by _____.
- xii.** Combination of all critical activities will formulate _____.

D) True or False (xiii – xv) (Note: Encircle the correct answer)

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- xiii.** LCM is always better than VAM and MMM.
(True) or (False)
- xiv.** Simplex is only for the minimization problem.
(True) or (False)
- xv.** Objective of an assignment problem is always to maximize profit.
(True) or (False)

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BM208.1 OPERATIONS RESEARCH

Date: 03/05/2018, Thursday

Time: 10.30 AM to 01:00 PM

Total Marks: 55

- Note:** - (i) All questions are compulsory.
(ii) This is a closed book examination.
(iii) There are two sections in the question paper. **Section – I** contains Q. No.: 1 of 15 marks and **Section – II** contains total five questions where in Q. No.: 2 to 5 are of 10 marks each and Q. 6 is a Case study of 15 marks.
(iv) **Section – I** should be answered in the given question paper and **Section – II** should be answered in a separate answer book.
(v) Maximum time allotted for **Section – II** is 150 minutes.
(vi) Figures on the right indicate marks for each question.

SECTION – II

- Q.2** Solve following LPP using graphical method. **10**
Max $Z = 5X + 8Y$
Subject to,
 $3X + 2Y \leq 36$
 $X + 2Y \leq 20$
 $3X + 4Y \leq 42$
 $X, Y \geq 0$

OR

- Q.2 [A]** Determine the optimal strategies for the following game. **[5] 10**

		Player B			
		B1	B2	B3	B4
Player A	A1	-5	2	0	7
	A2	5	6	4	8
	A3	4	0	2	-3

- [B]** Determine the optimal strategies for the following game. **[5]**

		Player B		
		B1	B2	B3
Player A	A1	-1	2	-2
	A2	6	4	-6

- Q.3** Solve following Transportation Problem using VAM. **10**

		Warehouse						Supply
		A	B	C	D	E	F	
Factory	1	9	12	9	6	9	10	5
	2	7	3	7	7	5	5	6
	3	6	5	9	11	3	11	2
	4	6	8	11	2	2	10	9
	Demand	4	4	6	2	4	2	

Q.4 [A] Solve Following Assignment Problem.

[5] **10**

Machine \ Job	Time (Hours)			
	J1	J2	J3	J4
M1	14	5	8	7
M2	2	12	6	5
M3	7	8	3	9
M4	2	4	6	10

[B] Solve Following Assignment Problem.

[5]

Machine \ Job	Time (Hours)			
	J1	J2	J3	J4
M1	11	14	6	0
M2	8	10	11	0
M3	9	12	7	0
M4	10	13	8	0

OR

Q.4 A small maintenance project consists of following jobs, whose precedence relationships are given below. **10**

Activity	A	B	C	D	E	F	G	H	I	J
Job	1-2	1-3	2-3	2-5	3-4	3-6	4-5	4-6	5-6	6-7
Duration (Days)	15	15	3	5	8	12	1	14	3	14

Draw Network Diagram, Calculate Total Float for each activity and identify critical path with total project duration.

Q.5 The annual requirement for a particular raw material is 2000 units, costing Rs. 1 each to manufacture. The ordering cost is Rs. 10 per order and the carrying cost is 16% of the cost of the product. Determine optimum inventory policy. **10**

Q.6 Case Study: **15**
H. T. Electric Company located in Chennai's Guindy Industrial Area, produces two products namely chandeliers and ceiling fans. Both these products require a two-step production process involving wiring and assembly. It takes about 20 hours to wire each chandelier and 30 hours to wire a ceiling fan. Final assembly of the chandeliers and fans requires 60 and 50 hours respectively. The production capability is such that only 120 hours of wiring time and 300 hours of assembly time are available. The profit obtained by the firm is Rs. 70 per unit of chandelier and Rs. 60 per unit of ceiling fan sold. Determine the optimal product-mix using graphical method.